

Class 10 NCERT Maths

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Chapter 9: Some Applications of Trigonometry

CHAPTER 9: SOME APPLICATIONS OF TRIGONOMETRY

A simplified, detailed guide in Hinglish

For students who missed Class 7-8-9

Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 IS CHAPTER MEIN KYA HAI?

Chapter 8 mein trigonometry seekhi. Ab use real-life mein lagayenge - HEIGHTS aur DISTANCES nikaalne ke liye.

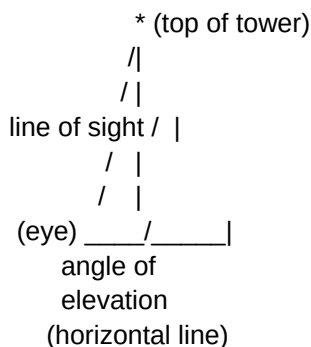
Bina pahaadi/building pe chadhe uski height nikaal sakte hain - sirf angle aur distance se! Ye magic trig se hota.

1.2 LINE OF SIGHT

LINE OF SIGHT = aankh se object tak ki seedhi line (jab hum kisi cheez ko dekhte hain).

1.3 ANGLE OF ELEVATION (Upar dekhna)

Jab object hamse UPAR ho (jaise building ka top), toh horizontal line se upar wala angle = ANGLE OF ELEVATION.



1.4 ANGLE OF DEPRESSION (Neeche dekhna)

Jab object hamse NEECHE ho (jaise cliff se boat dekhna),
toh horizontal line se neeche wala angle = ANGLE OF
DEPRESSION.

(eye)_____ horizontal
 \
 \ angle of
 \ depression
 \
 \ line of sight
 \
 * (boat below)

IMPORTANT FACT:

Angle of elevation (neeche se upar)
= Angle of depression (upar se neeche)
(Alternate angles - parallel horizontal lines)

SECTION 2: KEY APPROACH (HOW TO SOLVE)

2.1 STEPS TO SOLVE ANY PROBLEM

1. DIAGRAM banao (right triangle banta hai)
2. Known angle aur known side mark karo
3. Unknown (height/distance) ko x maan lo
4. Decide karo konsa ratio use karna hai:

$\sin = \text{Opp/Hyp}$ -> use jab Hyp involved
 $\cos = \text{Adj/Hyp}$ -> use jab Hyp involved
 $\tan = \text{Opp/Adj}$ -> MOST USED (height & base)

5. Equation banao aur solve karo

2.2 SPECIAL ANGLE VALUES (RECAP)

$\tan 30 = 1/\sqrt{3} = 0.577$
 $\tan 45 = 1$
 $\tan 60 = \sqrt{3} = 1.732$

(Height/distance problems mein zyadatar tan use hota)

SECTION 3: WORKED EXAMPLES

EXAMPLE 1: SIMPLE TOWER

"Ek tower ki base se 30 m door khade ho. Tower ke top ka angle of elevation 60 degree hai. Tower ki height?"

Diagram: right triangle

Base = 30 m, angle = 60, height = h (opposite)

Use tan (opp/adj):

$$\tan 60 = h / 30$$

$$\sqrt{3} = h / 30$$

$$h = 30 \times \sqrt{3} = 30 \times 1.732$$

$$h = 51.96 \text{ m}$$

Tower ki height = $30 \times \sqrt{3}$ m (approx 51.96 m)

EXAMPLE 2: FINDING DISTANCE

"Ek 15 m unchi building ke top se ek car ka angle of depression 45 degree hai. Car building se kitni door?"

Angle of depression = angle of elevation = 45 (alternate)
height = 15 m, distance = d

$$\tan 45 = 15 / d$$

$$1 = 15 / d$$

$$d = 15 \text{ m}$$

Car 15 m door hai.

EXAMPLE 3: LADDER PROBLEM

"Ek ladder wall se 60 degree angle banati hai. Ladder ka foot wall se 2.5 m door hai. Ladder ki length?"

Here: base (adjacent) = 2.5 m, ladder = hypotenuse = L
angle with ground = 60

$$\cos 60 = \text{adj} / \text{hyp} = 2.5 / L$$

$$1/2 = 2.5 / L$$

$$L = 5 \text{ m}$$

Ladder ki length = 5 m

EXAMPLE 4: TWO ANGLES (slightly harder)

"Ek tower ke base se kuch door se top ka angle of elevation 30 hai. 20 m aur paas aane par 60 ho jaata. Tower ki height nikaal."

Let height = h, near distance (from base) = x.

Far point = $(x + 20)$.

From near point: $\tan 60 = h/x \rightarrow h = x\sqrt{3} \dots(1)$

From far point: $\tan 30 = h/(x+20)$

$$1/\sqrt{3} = h/(x+20)$$

$$h\sqrt{3} = x + 20 \dots(2)$$

Put (1) in (2):

$$(x\sqrt{3})\sqrt{3} = x + 20$$

$$3x = x + 20$$

$$2x = 20$$

$$x = 10$$

$$h = x\sqrt{3} = 10\sqrt{3} = 17.32 \text{ m}$$

Tower height = $10\sqrt{3}$ m (approx 17.32 m)

Q AND A TIME

Q1. Angle of elevation aur angle of depression mein kya difference hai? Ek line mein bata.

Q2. "Tower ki base se 40 m door se top ka angle of elevation 45 hai. Tower ki height?"

Q3. "Ek 20 m unchi cliff ke top se boat ka angle of depression 30 hai. Boat kitni door hai?"
($\tan 30 = 1/\sqrt{3}$ use kar)

Q4. "Ek patang (kite) ki dori 100 m lambi hai aur ground se 30 degree angle banati hai. Patang ki height kitni hai?" ($\sin 30 = 1/2$ use kar)

Q5. Ek tower $50\sqrt{3}$ m unchi hai. Uski base se kuch door se top ka angle of elevation 60 hai. Door ki distance nikaal.

SUMMARY

1. Line of sight: aankh se object tak seedhi line.
2. Angle of ELEVATION: object upar ho (upar dekhna).

3. Angle of DEPRESSION: object neeche ho (neeche dekhna).
4. Elevation angle = Depression angle (alternate angles).
5. Solving steps:
 - Diagram banao
 - tan/sin/cos choose karo
 - Equation banao aur solve karo
6. tan zyadatar use hota hai (height aur base ka rishta).

TIPS FOR EXAM

1. DIAGRAM zaroor banao - aadha kaam wahin ho jaata hai.
2. Angle of depression ko triangle ke andar wale angle mein convert kar (alternate angle = elevation).
3. $\tan = \text{height/base}$ yaad rakh - 90% problems isi se.
4. Answer mein $\sqrt{3} = 1.732$, $\sqrt{2} = 1.414$ use karke decimal bhi likh sakte ho (agar poocha jaye).
5. Units (m, km) final answer mein likhna mat bhoolna.